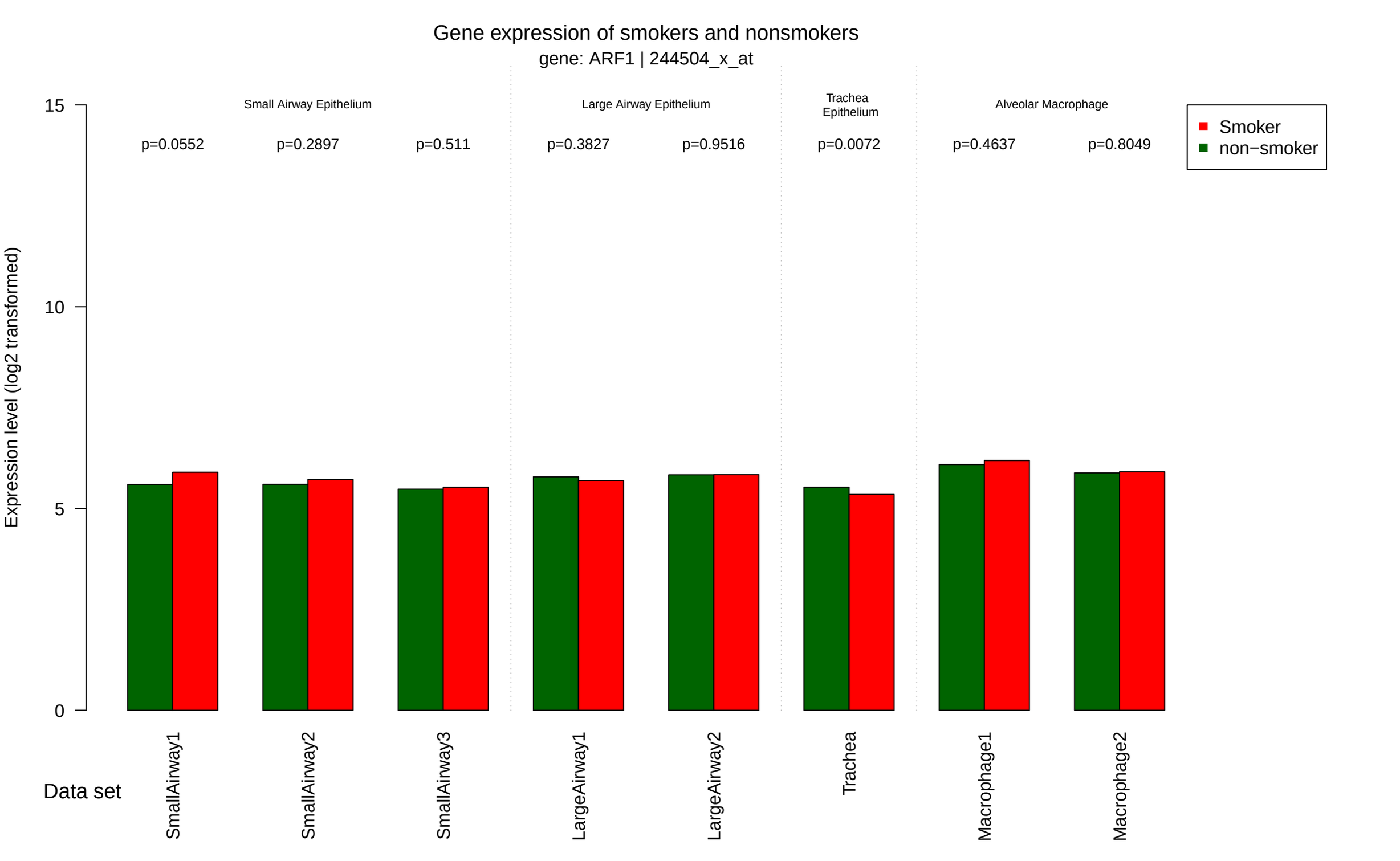
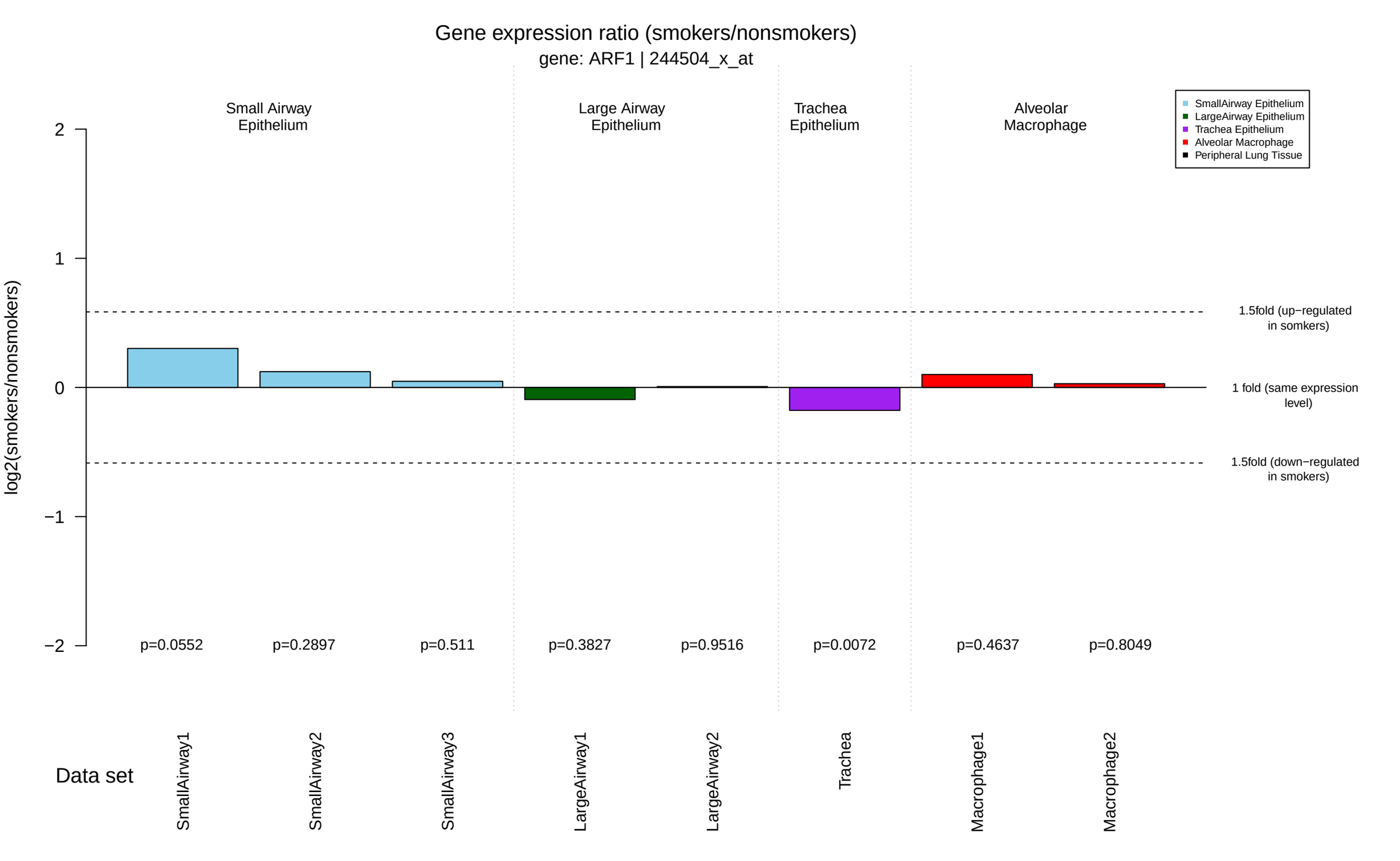
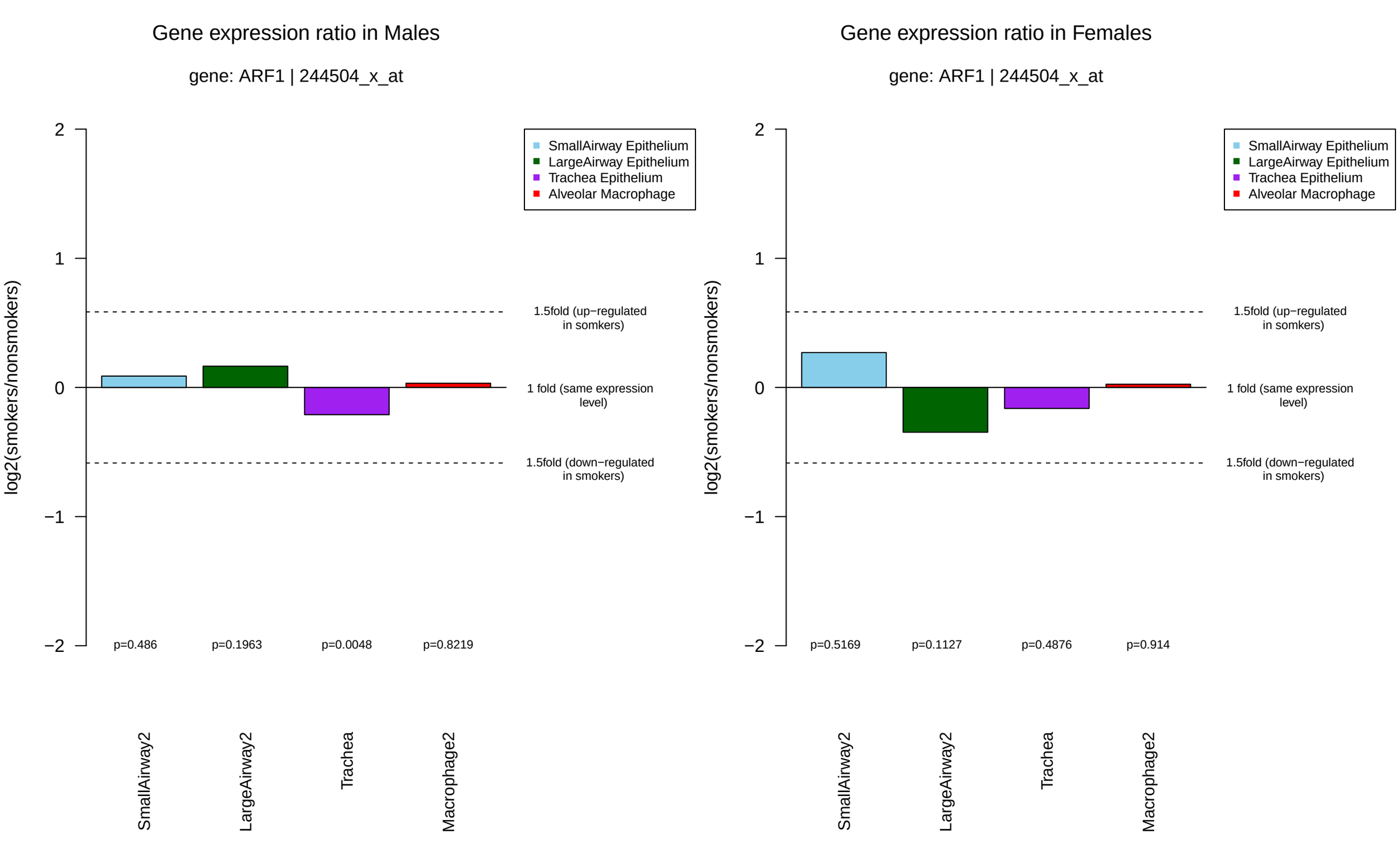
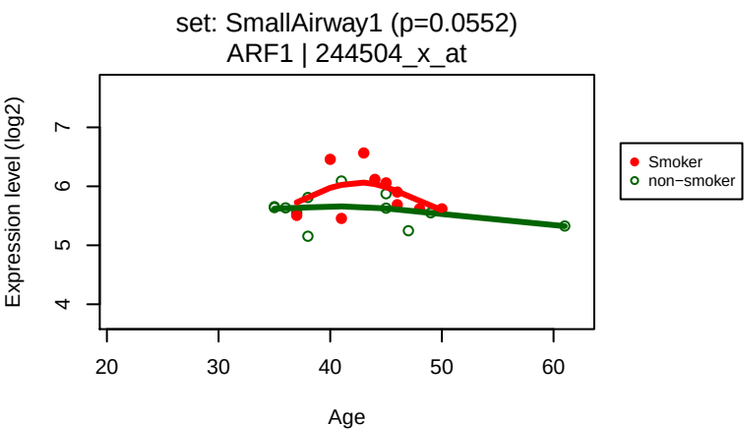
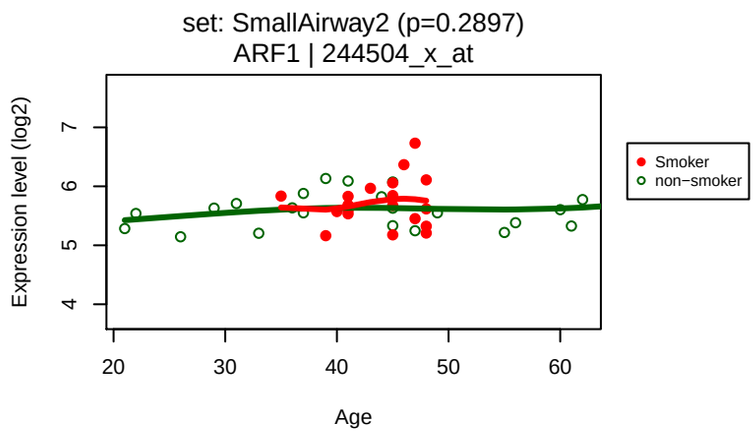
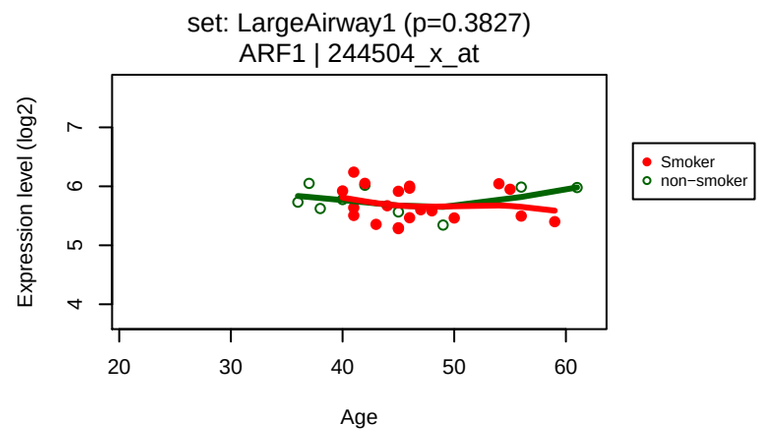

Gene expression of smokers and nonsmokers
gene: ARF1 | 244504_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0552
p=0.2897
p=0.511
p=0.3827
p=0.9516
p=0.0072
p=0.4637
p=0.8049
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

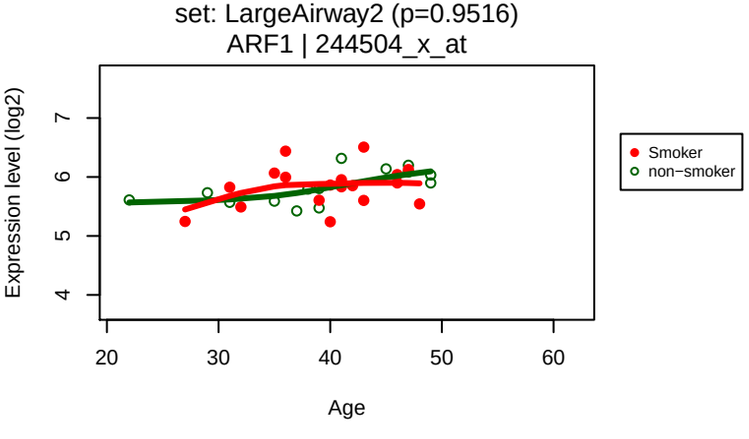

Gene expression ratio (smokers/nonsmokers)
gene: ARF1 | 244504_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0552
p=0.2897
p=0.511
p=0.3827
p=0.9516
p=0.0072
p=0.4637
p=0.8049
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

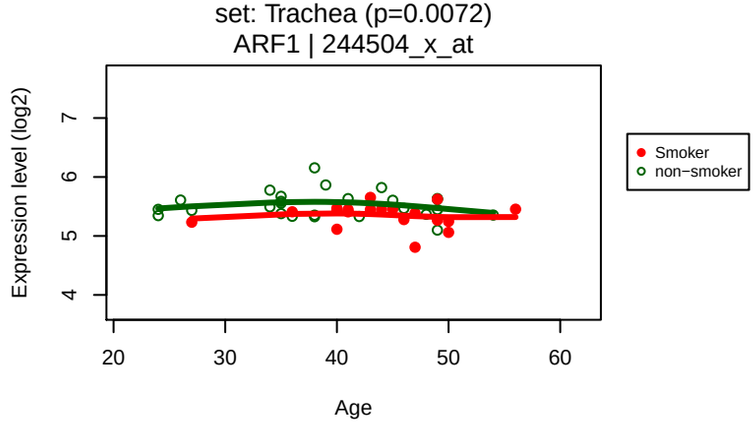

Gene expression ratio in Males
gene: ARF1 | 244504_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.486
p=0.1963
p=0.0048
p=0.8219
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARF1 | 244504_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5169
p=0.1127
p=0.4876
p=0.914
SmallAirway2
LargeAirway2
Trachea
Macrophage2

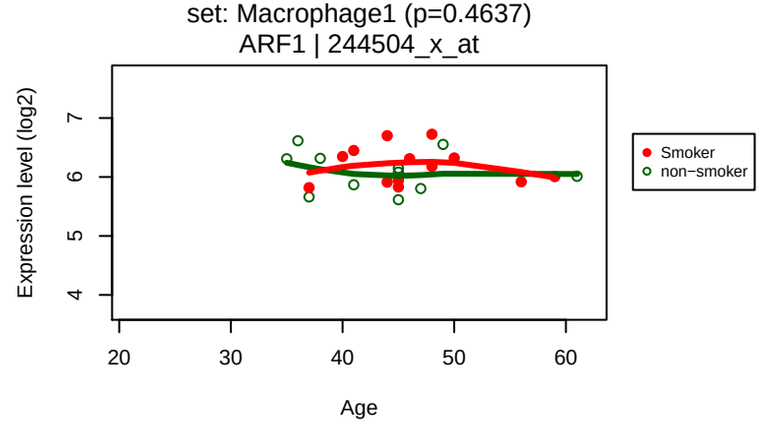

set: SmallAirway1 (p=0.0552)
ARF1 | 244504_x_at
Expression level (log2)
Age
Smoker
non−smoker

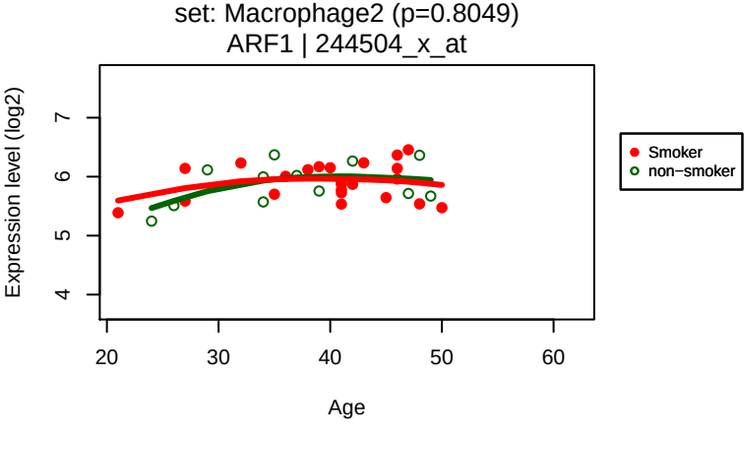

set: SmallAirway2 (p=0.2897)
ARF1 | 244504_x_at
Expression level (log2)
Age
Smoker
non−smoker

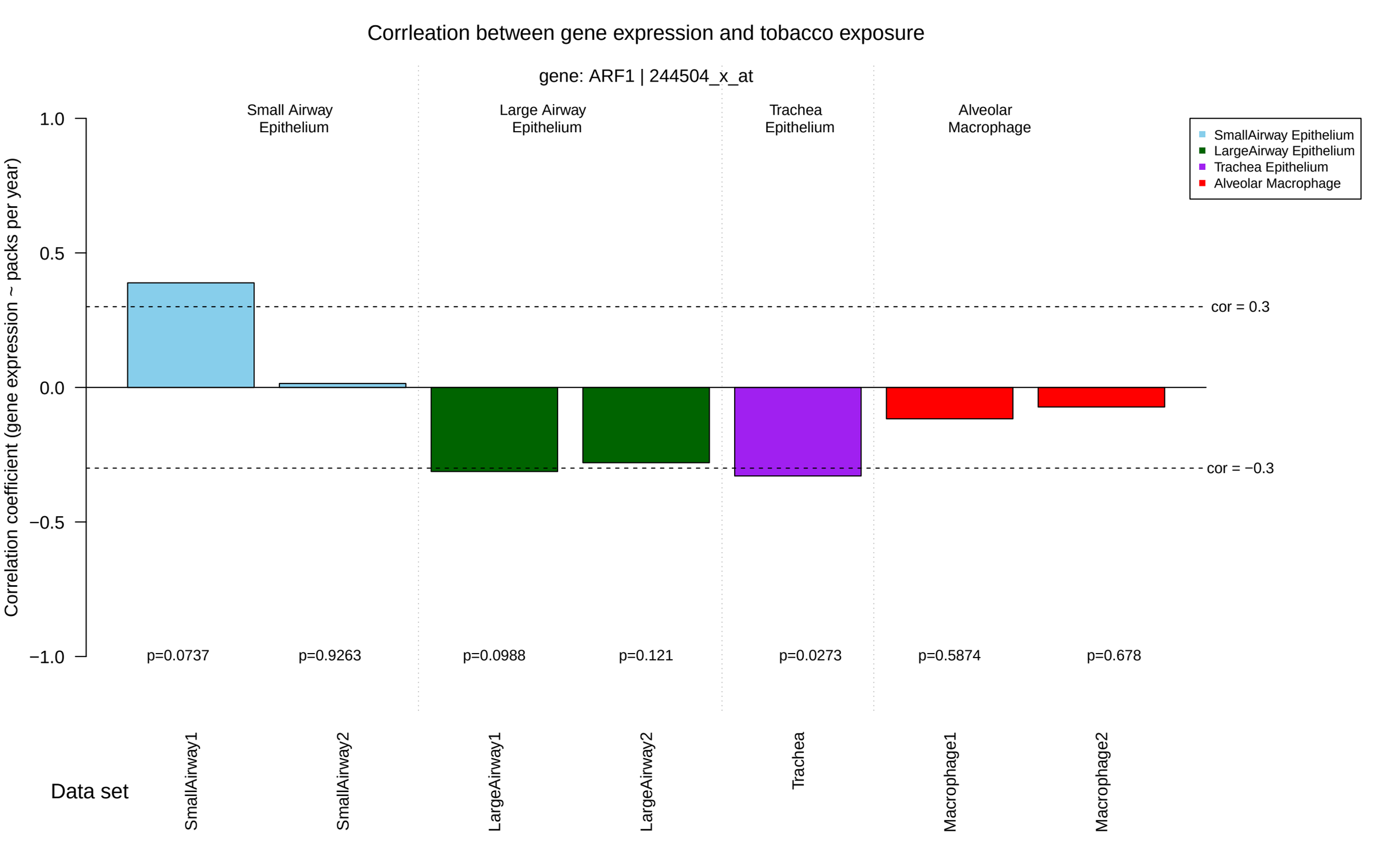

set: LargeAirway1 (p=0.3827)
ARF1 | 244504_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9516)
ARF1 | 244504_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0072)
ARF1 | 244504_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4637)
ARF1 | 244504_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8049)
ARF1 | 244504_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARF1 | 244504_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0737
p=0.9263
p=0.0988
p=0.121
p=0.0273
p=0.5874
p=0.678
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2